



TAVAONE EDUCATION · PINELLAS STEM RADIO LAB

BSA Radio Merit Badge

Session 1 — Radio Foundations

Safety · Fundamentals · Propagation · Regulation · Amateur Radio (Reqs 1–6, 8-A(1–5))

TavaOne Education · Pinellas STEM Radio Lab · Gulf Coast Contest Club



Today's Agenda

- Req 1 — Radio Safety (grounding, burns, Li-ion fires)
- Req 2 — Radio Fundamentals (EM spectrum chart, definitions)
- Req 3 — How Radio Waves Travel (propagation)
- Req 6 — Radio Regulation (FCC/ITU, call signs, phonetic alphabet)
- Req 8-A, Items 1–5 — Amateur Radio knowledge
- Req 9 — Introduction to career/hobby research assignment
- On-Air: Supervised voice contact at the club station

Requirement 1

Radio Safety — Grounding, Burns & Battery Fires



Req 1a — Grounding: Two Types of Circuits

DC Grounding

- Batteries and solar panels produce DC
- Negative terminal = ground reference
- All negative leads connect to a common ground point
- Prevents voltage differences and arcing between equipment

AC Grounding (Household Power)

- Household outlets are 120 V AC in the US
- The third (round) prong is the safety ground
- Never remove the third prong or use an ungrounded adapter
- Safety ground prevents the chassis from becoming energized



Req 1a — Antenna & Lightning Grounding

- ⚡ NEVER touch an antenna or feedline during a thunderstorm
- Disconnect feedlines from radio BEFORE the storm arrives
- Lightning arrestors divert strike energy to a ground rod — not through your radio
- Antennas build up static charge even without lightning → a bleeder resistor or ground strap bleeds it off safely
- Ground rod should be driven into moist earth, bonded to station ground bus
- If outdoors near an antenna in a storm → move indoors immediately



Req 1b — Electrical Burns: Prevention & Treatment

Prevention

- Always turn off and unplug before working inside equipment
- Capacitors hold charge after power is off — discharge them first
- One-hand rule: keep one hand in pocket when probing live circuits
- Never work on electrical equipment alone

Treatment

- FIRST: cut power before touching the victim
- Call 911 — even if person "seems okay"
- Cool running water on the burn — NOT ice
- Cover loosely with sterile dressing, do not pop blisters
- Internal damage is often far worse than external burns look



Req 1c — Lithium-Ion Battery Fire Prevention

- Li-ion batteries power phones, handheld radios, tablets, laptops
- THERMAL RUNAWAY: heat from one failing cell ignites adjacent cells — the fire spreads and re-ignites
- Never use a charger not designed for your specific battery
- Do not charge unattended overnight without a protected quality charger
- Swollen (puffy) battery = failing → stop using, dispose properly
- Never puncture, crush, or expose to temperatures above 140°F (60°C)
- Li-ion fire: do NOT use water → use Class D extinguisher or sand, move device outdoors, call 911



Quick Check

Your baofeng battery is puffy and the case feels warm. What do you do?

A) Charge it — the warmth means it needs more power

B) Continue using it, it's probably fine

C) Stop using it, do not charge it, dispose of it properly

D) Put it in the freezer to cool down

Answer: C — A swollen battery is failing. Stop using it immediately. Disposing properly means taking it to a battery recycling point, not the trash.

Requirement 2

Radio Fundamentals — Spectrum, Definitions & Discussion



Req 2a — The Electromagnetic Spectrum (300 kHz – 3 GHz)

- All radio waves are EM waves — same physics as light, just longer wavelength
- Frequency $\uparrow$ = Wavelength $\downarrow$ (they're inversely proportional)
- MF (300 kHz–3 MHz): AM broadcast, amateur 160m/80m, maritime
- HF (3–30 MHz): Shortwave, all ham HF bands, CB radio (27 MHz), WWV
- VHF (30–300 MHz): FM broadcast (88–108), TV, amateur 6m/2m, public safety
- UHF (300 MHz–3 GHz): TV, amateur 70cm, cellular, Wi-Fi 2.4 GHz, GPS, Bluetooth
- Microwave (3 GHz+): Wi-Fi 5 GHz/6 GHz, satellite TV, radar, 5G mmWave
- ACTIVITY: Draw your own chart — label all five bands and mark 8 radio services



Req 2b — Three Discussion Topics

What is radio? · Broadcast vs. Two-Way

- Radio: transmission of info using EM waves in the RF spectrum
- TX converts audio/data → RF; antenna radiates it; RX captures and decodes
- Broadcast: one-way, one-to-many (AM/FM, TV)
- Two-way: both parties transmit and receive (ham, walkie-talkie, police)

Commercial Broadcast vs. Hobby Radio

- Commercial: licensed for business, sells advertising, passive audience
- Amateur (hobby): non-commercial, licensed for personal use and experimentation
- Hams build equipment, experiment with antennas, provide emergency comms
- No selling ads, no playing music for an audience

Requirement 3

How Radio Waves Travel — Propagation



Req 3a — Three Propagation Modes

- GROUND WAVE: Follows the earth's surface → MF/AM broadcast → range: a few hundred miles
- SKY WAVE (Ionospheric): Signal goes up → hits the ionosphere → bends back down → HF can travel thousands of miles → explains why you hear faraway AM stations at night
- LINE-OF-SIGHT (Direct wave): VHF and above travel in straight lines → limited by terrain and antenna height → FM radio, TV, 2m amateur, cellular

- ACTIVITY: Sketch cross-section of earth, show all three paths with arrows
- Label: transmitter, ionosphere layer, skip zone, receiver at distant skip distance



Req 3b & 3c — WWV/WWVH and DX vs. Local

WWV and WWVH — Band Condition Indicators

- NIST time-standard stations: WWV (Fort Collins CO), WWVH (Kauai HI)
- Broadcast on 2.5, 5, 10, 15, 20 MHz — 24/7
- If you can hear WWV on 20m → 20m is OPEN to Colorado
- Weak signal → marginal conditions
- Can't hear at all → band may be closed (absorbed)
- Propagation forecast voiced at :18 each hour

DX vs. Local Stations

- LOCAL: close enough for reliable ground wave or LOS reception → typically within 25–100 miles, strong and consistent
- DX: "distance" — distant station reached by sky wave → signal fades in and out (QSB) → may be thousands of miles away
- DX-ing is a major ham pursuit: rare countries, expeditions to remote islands, contests to work as many countries as possible
- A "DX station" calling CQ on HF is usually someone YOU want to contact

Requirement 6

Radio Regulation — FCC, ITU, Call Signs & Phonetic Alphabet



Req 6a — FCC vs. ITU

FCC (Federal Communications Commission)

- US government agency created by Communications Act of 1934
- Regulates all radio, TV, wire, satellite, and cable in the US
- Issues licenses, enforces rules, allocates US frequencies
- Runs the amateur service via Volunteer Examiner Coordinators (VECs)
- Investigates interference complaints

ITU (International Telecommunication Union)

- UN agency — coordinates global use of radio spectrum and satellite orbits
- Issues the Radio Regulations — an international treaty all nations agree to follow
- Divides the world into 3 Regions; assigns frequency bands to services in each
- FCC rules must comply with ITU at the international level
- Simple: ITU sets global rules → FCC enforces them in the US



Req 6b — Call Signs

- AMATEUR: Prefix + Number + Suffix → e.g., W4GGJ
- US prefixes: A, K, N, W — ITU-assigned
- Number = call district (0–9 in continental US)
- Must ID at end of every transmission and at least every 10 minutes
- Your call sign identifies you — not your radio

- BROADCAST: W____ for stations east of Mississippi, K____ west (with historic exceptions)
- Must ID at top of each hour and at sign-on/sign-off
- TV call signs use -TV suffix (WFLA-TV)
- If there is a tie for "first use" — historical call signs may break the east/west rule (e.g., KDKA in Pittsburgh)



Req 6c — NATO Phonetic Alphabet

- A—Alfa · B—Bravo · C—Charlie · D—Delta · E—Echo · F—Foxtrot · G—Golf
 - H—Hotel · I—India · J—Juliett · K—Kilo · L—Lima · M—Mike
 - N—November · O—Oscar · P—Papa · Q—Quebec · R—Romeo · S—Sierra · T—Tango
 - U—Uniform · V—Victor · W—Whiskey · X—X-ray · Y—Yankee · Z—Zulu
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- PRACTICE: Spell your first name using phonetic alphabet — say it out loud
 - PRACTICE: Spell your troop number phonetically
 - WHY: B/D/E/G/P/T/V all sound alike on a noisy radio channel — these words are chosen for maximum distinctiveness



Quick Check

A ham must give their call sign: ____

A) Only at the beginning of each transmission

B) At the end of every transmission and at least every 10 minutes

C) Only when asked by another station

D) Only when transmitting on HF

Answer: B — End of every transmission AND at least every 10 minutes during a QSO (contact). This is FCC Part 97 rule.

Requirement 8-A

Amateur Radio — Items 1 through 5



8-A(1) — Why the FCC Has an Amateur Radio Service

- Emergency communications: hams provide comms when cell/internet fails in disasters (ARES, RACES)
- Advancement of the radio art: hams experiment and innovate — some innovations later go commercial
- International goodwill: connecting people across borders without commercial or political purpose
- Pool of trained operators: national asset in wartime or national emergency
- What licensed hams can do: voice, digital, satellite (AMSAT), meteor scatter, moonbounce (EME), POTA/SOTA, APRS tracking, public service events, emergency communication



8-A(2) — License Classes and VE Exams

- TECHNICIAN (Element 2): 35 Qs from 423-Q pool, pass 26/35 (74%). All VHF/UHF; limited HF.
- GENERAL (Elements 2+3): 35 Qs from 432-Q pool, pass 26/35. Technician privileges + most HF worldwide.
- AMATEUR EXTRA (Elements 2+3+4): 50 Qs from 622-Q pool, pass 37/50. All amateur privileges; exclusive sub-bands; premium callsigns.

- VE Exams: administered by Volunteer Examiners (VEs) — licensed hams accredited through Volunteer Examiner Coordinators (VECs) like ARRL-VEC, W5YI-VEC.
- FCC does NOT administer the exams directly.
- Exam fee: ~\$15 (VEC session) + \$35 FCC application fee after passing.



8-A(3) — Q Signals and Amateur Terms

- QSO — A contact / "Can you communicate with ___?"
- QTH — Location / "My location is ____."
- QSL — Acknowledgment (also a confirmation card) / "I confirm the contact."
- QRM — Interference from other stations / "I am being interfered with."
- QRN — Static / "I am troubled by static."
- QRZ — "Who is calling me?" / Used to ask who is there
- QSB — Fading signals / "Your signals are fading."
- CQ — General call: "Is anyone listening? I want to make a contact."
- 73 — Best regards (standard friendly sign-off)
- RST — Signal report: Readability (1–5) / Strength (1–9) / Tone (1–9 for CW)



8-A(4) — Emergency Calls on Voice and CW

Voice — MAYDAY

- MAYDAY MAYDAY MAYDAY
- [your call sign]
- [your location]
- [nature of emergency]
- [number of people]
- [any other critical info]
- Repeat as needed on 146.520 MHz (2m simplex), 14.300 MHz (HF maritime mobile)

Morse Code — SOS

- ... — — — ... (SOS)
- SOS is NOT an abbreviation — it is one unbroken character: 3 dits, 3 dahs, 3 dits
- Follow with DE [your callsign] — [location] — [situation]
- Sent continuously until help responds or you must stop



8-A(5) — Transceivers and Repeaters

- **HANDHELD** (HT — Handie-Talkie): battery-powered, portable, ~5 W, VHF/UHF, great for local use and public service events. Examples: Yaesu FT-60, Baofeng UV-5R.
- **MOBILE**: vehicle-installed, 12 V power, 25–50 W on VHF/UHF, up to 100 W HF, larger antenna = more range. Examples: Icom IC-7100, Yaesu FT-891.
- **BASE STATION**: fixed location, AC-powered, 100–1,500 W HF, large permanent antennas, maximum range. Examples: Icom IC-7300, Kenwood TS-890.
- **REPEATER**: Listens on the INPUT frequency, retransmits on the OUTPUT frequency from a high location at higher power. Extends HT range from ~5 miles to 50+ miles.
- Your radio needs: correct offset (e.g., -600 kHz on 2m) + CTCSS tone to open the repeater.
- Many repeaters link via internet (IRLP, EchoLink, AllStar) for worldwide reach.



Quick Check

What does a repeater's "offset" mean?

- A) The time delay before the repeater responds
- B) The difference between the repeater's input and output frequencies
- C) The CTCSS tone code required to open the repeater
- D) The maximum range of the repeater system

Answer: B — The offset is the frequency difference between where the repeater listens (input) and where it retransmits (output). On 2m this is typically -600 kHz.

Requirement 9

Career & Hobby Exploration — Homework Assignment



Req 9 — Your Assignment (due before Session 2)

Option 9a — Radio Career

- Research ONE career related to radio
- Must cover: training, education needed, costs, job prospects, salary, duties, advancement
- Good choices: broadcast engineer, RF engineer, antenna tech, air traffic controller, emergency manager, radio astronomer
- Discuss findings with counselor in Session 2

Option 9b — Ham Radio Hobby or Volunteer

- Research amateur radio hobby OR a volunteer role (ARES, RACES, public service)
- Must cover: licensing and training, startup costs, supporting organizations
- Discuss short-term AND long-term goals you might have
- Interview a ham operator if possible

On-Air Time

Supervised Voice Contact — Req 8-A(6) Preparation



Your First On-Air Contact — Voice Operating Procedure

- 1. LISTEN FIRST: Is the frequency already in use? "Is this frequency in use?" — wait 5 seconds.
 - 2. CALL CQ: "CQ CQ CQ, this is [callsign], CQ and standing by."
 - 3. ANSWER: "[Their callsign], this is [your callsign] — or — just [your callsign]"
 - 4. EXCHANGE: Give name, QTH (location), and RST signal report
 - 5. LOG IT: Date, UTC time, their call sign, band/frequency, mode, RST sent, RST received, their QTH
 - 6. SIGN OFF: "73, this is [callsign] clear and listening."
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- RST reminder: "You are 5-9" = perfectly readable, extremely strong signal. "You are 5-5" = perfectly readable, middling strength. Most contacts get 5-9 if they're clear.



Session 1 Complete!

What to do before Session 2:

- Complete your Req 9 research (career OR hobby/volunteer path)
- Review the Q signals (QSO, QTH, QSL, QRM, QRN, QRZ, QSB)
- Practice the phonetic alphabet until you can spell your name without looking
- Bring this student pack, your blue card, and your Req 9 notes to Session 2